ERMITTLUNGSAKTE · CASE FILE REFERENCE · EXCEL FOR THE WEB · GERMAN FORMULAS

TATORT: EXCEL

A Forensic Guide to Data Cleaning. Function names are given as German Excel shows them, with the English name in brackets. Formulas use the `;` separator; if your Excel uses `,` (German language, US regional format), swap them, but not inside quotation marks.

THE INVESTIGATION

Step	Action	Why it matters
0 · Secure the Scene	Duplicate the raw sheet and work only on the copy. AutoSave is always on! Start a case log.	The original is your evidence. If anything goes wrong, you can always go back.
1 · Survey the Scene	Scroll everything. Freeze panes, turn on filters, open every drop-down. Note clues.	You can't fix what you haven't noticed. Look before you touch.
2 · Open the Case File	Write the order of operations. Log every change: what, why, how verified.	A plan makes the work repeatable, for colleagues and for future you.
3 · Clear the Scene	Delete junk rows above the header, irrelevant columns and exact duplicates.	Less to clean, fewer distractions.
4 · Process the Evidence	Fix spaces & hidden characters, data types, dates, case, missing-value placeholders.	Hidden problems break sorting, lookups, math and de-duplication.
5 · Identify the Suspects	Split combined columns, unmask aliases with a reference table, add validation.	One fact per column; one spelling per category.
6 · Cross-Examine	Remove duplicates again. Pivot tables, sorting and conditional formatting.	Confirms the cleaned data tells a consistent story.
7 · Close the Case	Save a clean copy, download a CSV, update the data dictionary, keep the case log.	Clean data is only useful if others can find, trust and reuse it.

WHERE TO FIND IT (EXCEL FOR THE WEB, GERMAN)

Tool	Location in German Excel (English)
Open a shared file	Open the link → **Datei** (File) → Create a Copy → Create a copy online. Your copy lives in your OneDrive
Open your own file	office.com → Excel → Upload a file
Copy a sheet	Right-click the sheet tab → Duplicate
Freeze header row	**Ansicht → Fenster fixieren → Oberste Zeile fixieren** (View → Freeze Panes → Freeze Top Row)
Filter	**Daten → Filter** (Data → Filter)
Make a Table	Click in the data → **Einfügen → Tabelle** (Insert → Table). New formula columns fill down by themselves
Find blanks	Filter a column → tick only **(Leer)** (Blanks); count with `=ANZAHLLEEREZELLEN(Bereich)`
Find & Replace	**Start → Suchen und Auswählen → Ersetzen** (Home → Find & Select → Replace), **Strg/⌘ + H**
Remove Duplicates	**Daten → Duplikate entfernen** (Data → Remove Duplicates)
Highlight duplicates	**Start → Bedingte Formatierung → Regeln zum Hervorheben von Zellen → Doppelte Werte**
Split a column	`TEXTVOR` / `TEXTNACH` / `TEXTTEILEN`, or search for *Text in Spalten* (Text to Columns)
Paste as values	Copy, then **Strg/⌘ + Umschalt + V** (Ctrl/⌘ + Shift + V)
Numbers stored as text	Helper column with `=WERT()` (VALUE), then paste values
Restrict entries	**Daten → Datenüberprüfung** (Data → Data Validation)
Spot-check	**Einfügen → PivotTable**; **Start → Bedingte Formatierung**

Tool	Location in German Excel (English)
Save a copy	**Datei** → Create a Copy → Create a copy online
Download a CSV	**Datei** → Export → Download as CSV UTF-8
Roll back a disaster	**Datei → Versionsverlauf** (File → Version History)
Can't find a button?	Type its German or English name in the search box at the top of Excel

Number formats: *General* is **Standard**; *Short Date* is **Datum, kurz**. TRUE/FALSE show as **WAHR/FALSCH**. On German keyboards Ctrl is **Strg**.

HANDY FUNCTIONS

German (English)	What it does	Example from the Tatort case
`GLÄTTEN` (TRIM)	Removes leading, trailing and double spaces	`=GLÄTTEN(B2)`
`SÄUBERN` (CLEAN)	Removes non-printing characters (line breaks, …)	`=GLÄTTEN(SÄUBERN(B2))`
`WECHSELN` (SUBSTITUTE)	Replaces specific text inside a cell	`=WECHSELN(G2; ZEICHEN(10); ", ")`
`UNIZEICHEN` (UNICHAR)	Any character by its Unicode number	`=GLÄTTEN(WECHSELN(B2; UNIZEICHEN(160); " "))`
`LÄNGE` (LEN)	Counts characters, including invisible ones	`=LÄNGE(B2)`
`ISTZAHL` / `ISTTEXT` (ISNUMBER / ISTEXT)	Reveals a cell's real data type	`=ISTZAHL(A2)`
`LET` (LET)	Gives a value a short name inside a formula	`=LET(d; [@Erstausstrahlung]; WENN(ISTZAHL(d); d; …))`
`WERT` (VALUE)	Turns number-looking text into a number	`=WERT(WECHSELN(A2; "#"; ""))`
`DATUM` (DATE)	Builds a real date from year, month, day	`=DATUM(RECHTS(D2;4); TEIL(D2;4;2); LINKS(D2;2))`
`TEXT` (TEXT)	Formats a number/date as text	`=TEXT(D2; "JJJJ-MM-TT")` (German format codes!)
`GROSS` / `KLEIN` / `GROSS2` (UPPER / LOWER / PROPER)	Changes capitalisation	`=GROSS2(B2)` (check the result!)
`IDENTISCH` (EXACT)	Case-sensitive comparison	`=IDENTISCH(B2; GROSS(B2))` flags ALL CAPS
`LINKS` / `RECHTS` / `TEIL` (LEFT / RIGHT / MID)	Pulls characters from a position	`=LINKS(E2; FINDEN("("; E2) - 1)`
`FINDEN` / `SUCHEN` (FIND / SEARCH)	Position of text (case-sensitive / not)	`=FINDEN("("; E2)`
`TEXTVOR` / `TEXTNACH` (TEXTBEFORE / TEXTAFTER)	Text before/after a delimiter	`=TEXTVOR(C2; "/"; ; ; ; C2)`
`TEXTTEILEN` (TEXTSPLIT)	Splits into several cells	`=TEXTTEILEN(C2; "/")`
`TEXTKETTE` / `&` / `TEXTVERKETTEN` (CONCAT / TEXTJOIN)	Combines text	`=A2 & " – " & B2`
`WENN` / `ODER` (IF / OR)	Tests a condition	`=WENN(ODER(G2="n/a"; G2="-"); ""; G2)`
`WENNFEHLER` (IFERROR)	Fallback when a formula errors	`=WENNFEHLER(LINKS(E2; FINDEN("(";E2)-1); E2)`
`XVERWEIS` / `SVERWEIS` (XLOOKUP / VLOOKUP)	Looks up a value in a reference table	`=XVERWEIS(C2; Known_Aliases!A:A; Known_Aliases!B:B; "CHECK")`
`ZÄHLENWENN` (COUNTIF)	Counts matches: finds duplicates	`=ZÄHLENWENN(A:A; A2) > 1`
`RUNDEN` / `AUFRUNDEN` / `ABRUNDEN` (ROUND / ROUNDUP / ROUNDDOWN)	Changes the stored value, not just the display	`=RUNDEN(H2; 0)`

KNOWN TRICKS OF THE TRADE

CAREFUL

GLÄTTEN and SÄUBERN miss the non-breaking space (character 160). Use `WECHSELN(…; UNIZEICHEN(160); " ")` first. `UNIZEICHEN` means the same thing everywhere.

CAREFUL

Find & Replace replaces text inside cells (`-` in *Schulze-Rohr*!) and treats `?` and `*` as wildcards. For placeholders, use `=WENN(ODER(G2="n/a"; G2="-"); ""; G2)` instead.

CAREFUL

`DATWERT` (DATEVALUE) follows your regional settings. `DATUM(Jahr; Monat; Tag)` never guesses, so build dates explicitly.

CAREFUL

`GROSS2` (PROPER) doesn't know German capitalisation (*Taxi nach Leipzig*) and breaks genuine acronyms (*KI*). Flag, then review.

CLUE

Same value ≠ duplicate. Three different episodes are called *Die Abrechnung*. Decide which columns define "the same record".

CLUE

Helper-column routine: add a column to the Table → formula (it fills down) → check (`LÄNGE` , filters, counts) → keep it beside the original. At the very end, in a copy: Paste Values over the original, delete helpers, log it.

CLUE

In a Table, `[@Titel]` = this row's `Titel` ; `[Titel]` = the whole column. Click a cell while writing a formula and Excel writes the reference for you.

CLUE

Clean, then de-duplicate again. Near-duplicates only match once spaces and spellings are fixed.